Di Wu

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Education

- **University of Science and Technology of China [🌐]** Hefei, China
  B.S. in Applied Physics Sep. 2021 - July 2025
  GPA: 3.91/4.30
  - **Relevant Coursework** :

Computer Architecture H	*93/100*	Computer Organization	*96/100*
Computer Programming A	*95/100*	Computational Method	*95/100*
Computer Network	*87/100*	Operation System	*85/100*
Linear Algebra	*100/100*	Artificial Intelligence Foundation	*93/100*
Probability Theory & Math. Statistics	*96/100*	Mathematical Analysis	*93/100*

- **University of Toronto[🌐]** Toronto, Canada
  Ph.D. in Computer Science, supervised by Prof. Qizhen Zhang Sep. 2025 - Dec. 2030 (expected)

Research Experience

- **Chair, the System and Networking Group, UofT** April. 2025 - Present
  Supervisor: Prof. **Qizhen Zhang**
  Project I: ***Massive Communication Framework in GPU for Database***
  - Conducted analysis of current GPU communication frameworks, and explored advanced technique.
  - Designed and implemented the new framework with high-performance and optimization support.

  Project II: ***GPU Database Benchmark***

- **Chair, the System and Networking Group, NUS** July 2024 - Jan. 2025
  Supervisor: Prof. **Bingsheng He** and Prof. **Yao Chen**
  Project I: ***Content Addressable Memory on FPGA***
  - Conducted analysis of FPGA-based CAM, enhancing the theoretical foundation for this DSA design.
  - Proposed a configurable DSP-based CAM architecture with high-performance and multi-query support.
  - Implemented an extensible CAM kernel with a hybrid HLS and RTL architecture on FPGA.
  - Submitted a research paper detailing the design to DAC'25.

- **Research Assistant, Institute of ASIC, USTC** June 2023 - June 2024
  Supervisor: Prof. **Lei Zhao**
  Project I: ***Test and analysis of a SerDes prototype chip based on 180 nm CMOS process***
  - Designed an efficient testing scheme tailored to the chip's specific structure and key testing metrics.
  - Investigated chip configurations and built test code using Verilog HDL.
  - Utilized FPGA to test the relevant index shortness, providing data to enhance the feed-through design.

  Project II: ***Simulation of the chip wiring S parameters***
  - Built the simulation models for chip wiring based on the encapsulated drawings.
  - Determined the S parameters to accommodate the shape material wiring, providing important information for the post-simulation circuit analysis.

Publications

C=Conference, J=Journal, O=Oral Presentation, S=In Submission, T=Thesis

[S.1] **Di Wu**, Hongshi Tan, Hanzhang Yang, Alireza Shateri, Bingsheng He, Qizhen Zhang. "**Magi: A Communication Framework for Data Processing in Massive GPU Infrastructures**," in *2026 VLDB*.

[C.1] Yao Chen, Feng Yu, **Di Wu**, Weng-Fai Wong and Bingsheng He. "**Configurable DSP-Based CAM Architecture for Data-Intensive Applications on FPGAs**," in *2025 62th Design Automation Conference (DAC)*.

[O.1] Jiacheng Guo, Jiajun Qin, Lei Zhao, **Wu Di**, Xinyu Bin, "**A 4×6.25-Gbs Serial Link Transmitter Core in 0.18-μm CMOS for high-speed front-end ASICs**," in *2024 24th IEEE Real Time Conference*.

[J.1] Yuheng Wu, Xiangdong Kong, Yechao Su, Jiankang Zhao, Yiling Ma, Tongzheng Ji, **Di Wu**, Junyang Meng, Yan Liu, Zhigang Geng, Jie Zeng, "**Thiol Ligand-Modified Au for Highly Efficient Electroreduction of Nitrate to Ammonia**," in *Precis. Chem. 2024, 2, 3, 112–119*.

TA Experience

- **University of Toronto**
  CSC258H1F Computer Organization Sep.2025 - Dec.2025

Skills & Interests

- **Programming Languages:** C/C++, Verilog/System Verilog, Python, LaTeX
- **Developer Tools:** Vivado, Vitis, Nsight
- **Technologies/Frameworks:** Linux, Git, HLS, CUDA, Pytorch
- **Interests:** Students Chorus Member, Piano